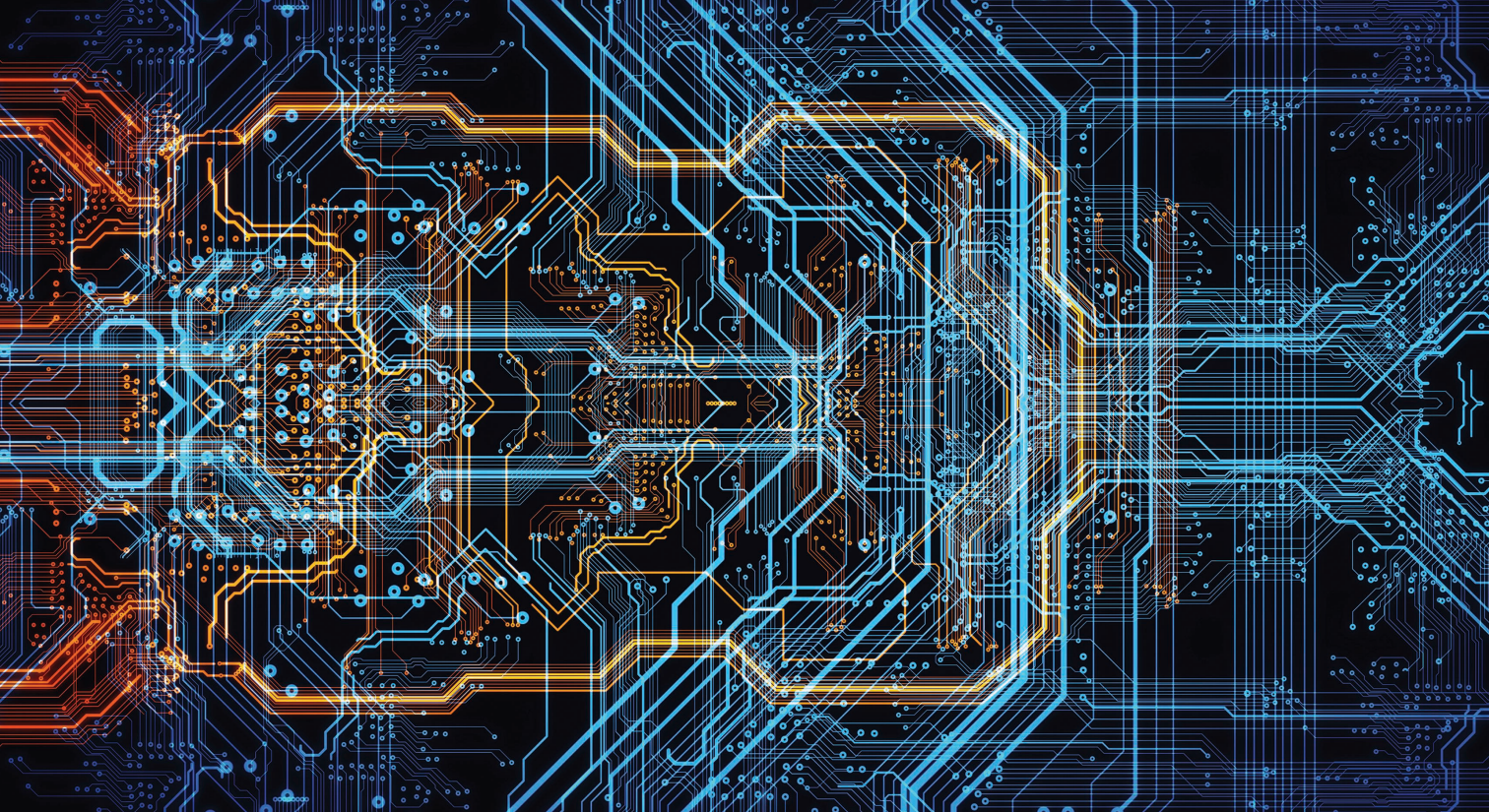




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High-Power-Density GaN-Based Converters

Thermal management considerations

by Edward A. Jones and Michael de Rooij

Increasing the power density of the latest generation of gallium nitride (GaN) transistor-based power converters, where smaller die sizes and chip-scale packages are deployed, can be achieved by adding a simple thermal solution that yields systems that raise the bar and set the standard for power density. Heat extraction from a small GaN transistor, or integrated circuit, can be as effective as, if not superior to, removing the heat from a bulky Si MOSFET when employing a good thermal design. This article presents a simple-to-implement thermal solution suitable for chip-scale GaN transistors. Furthermore, the assembly criteria and design boundaries are given with a derivation of the thermal models together with experimental validation.

Thermal Resistances in a Chip-Scale Package

When flipped, the back surface, known as the *case*, of flip-mounted chip-scale devices, such as land grid array (LGA) and ball grid array (BGA) transistors, becomes the top side. These wafer-level chip-scale packages (WLCSs) are thermally similar to other “flipped” device packages, for instance, DirectFET MOSFETs [1].

Figure 1 shows a cross-sectional diagram of a WLCS GaN transistor that has two distinct thermal-heat dissipation paths:

- 1) The first heat path is down through the BGA/LGA pads, solder joints, and into the printed circuit board (PCB). The heat may then be dissipated directly to ambient from the PCB, or through a heatsink on the opposite side of the PCB using thermal vias.
- 2) The second heat path is up through the top side of the die. If a heatsink is used, the heat passes through

a thermal interface material (TIM) and through the top-side heatsink to ambient. Absent a heatsink, only a small portion of heat can flow through the top of the die and by means of radiation and convection only.

Figure 2 illustrates the thermal resistances for the chip-scale packaged GaN transistor for the two examples given in Figure 1. The GaN transistor is mounted with the active device region facing the PCB and separated by the solder bumps/bars that form the device terminals. $R_{\theta JC}$ is defined as the thermal resistance from the active area (junction) of the die to the case and $R_{\theta JB}$ as the thermal resistance from junction to the board, which includes the solder bumps/bars to the copper traces on the PCB.

These thermal resistances, $R_{\theta JB}$ and $R_{\theta JC}$, for a chip-scale package are analogous to the thermal resistances, $R_{\theta JC(\text{bottom})}$ and $R_{\theta JC(\text{top})}$, for a lead-frame package, respectively. However, in contrast to a lead-frame package, both the top and bottom sides of a chip-scale GaN transistor are effective cooling paths.

Junction-to-Ambient Thermal Resistance

The overall thermal resistance between the junction and ambient, $R_{\theta JA}$, is often listed in device data sheets standardized to a specific PCB area (typically 1 in²) with no heatsink or forced air flow. This includes the PCB-to-ambient thermal resistance, $R_{\theta BA}$, as well as some direct case-to-ambient thermal resistance, $R_{\theta CA}$. However, the actual thermal resistance between junction and ambient will depend on the design of the converter, including PCB size and layout, air flow, and attachment of a heatsink. If the board-to-ambient and

case-to-ambient thermal resistances are very low, then the minimum possible junction-to-ambient thermal resistance is equivalent to $R_{\theta JC}$ in parallel with $R_{\theta JB}$.

Figure 3(a) shows the thermal resistances for some example Si MOSFETs and chip-scale GaN transistors. As shown, the bottom-cooling thermal resistance, $R_{\theta JB}$, follows an inverse trend with the device size. However, the top-cooling thermal resistance, $R_{\theta JC}$, is much lower for GaN transistors due to the chip-scale package. Figure 3(b) demonstrates that a chip-scale GaN transistor can achieve much lower $R_{\theta JA}$ with an effective

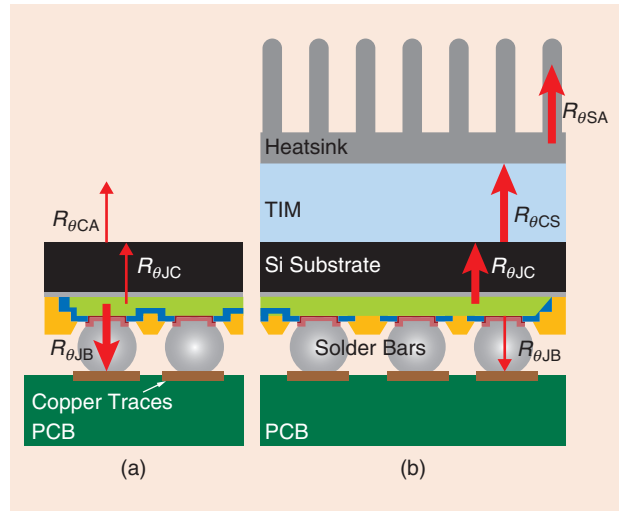


FIG 1 The cross section of a chip-scale packaged GaN transistor, highlighting the dominant heat flow paths from the top and bottom of the die. The diagrams illustrate both (a) without a heatsink and (b) with a heatsink attached to the top of the transistor. (Source: EPC; used with permission.)

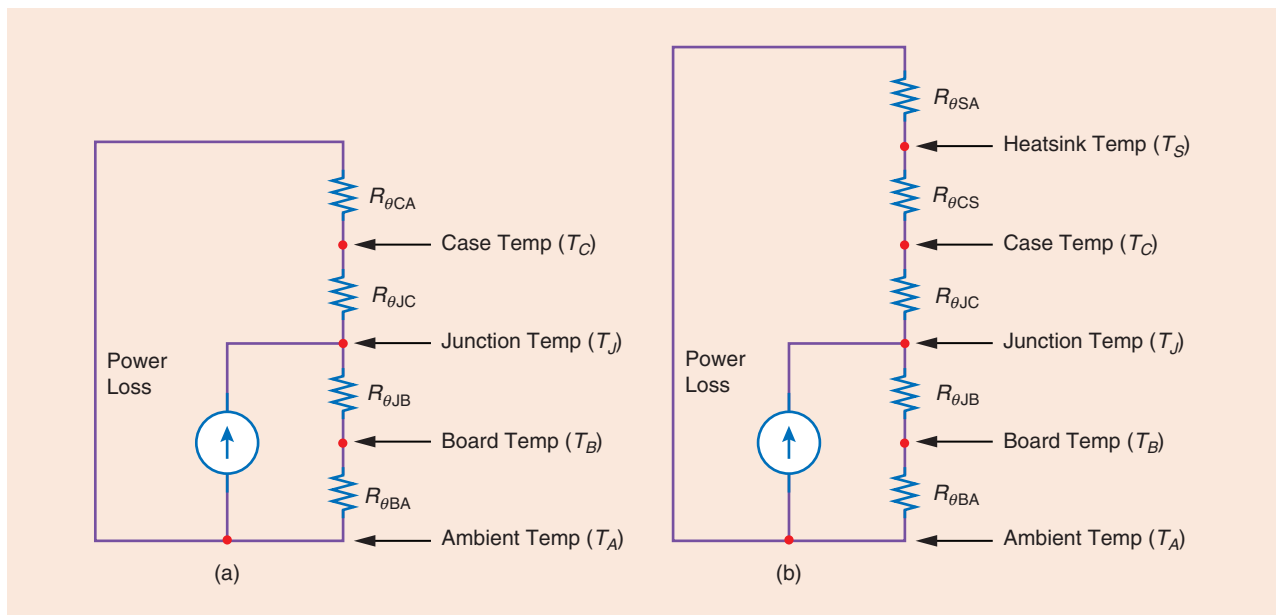


FIG 2 The steady-state thermal resistance schematic models for the physical device structures given in Figure 1. (a) Without a heatsink, and (b) with a heatsink attached to the top of the device. (Source: EPC; used with permission.)

heatsink connection on the top side [2]. Some design examples for top-mounted heatsinks will be discussed later in this article.

Improving Thermal Performance With a Heatsink

As with most converter designs, the thermal performance of GaN transistors can be improved by attaching a heatsink, reducing the thermal resistance between the junction and ambient air. This section discusses the selection of key thermal design components, as well as heatsink attachment schemes for both bottom-side and top-side or six-sided cooling.

Selection of Heatsink and TIM

The two primary components of a thermal design are the heatsink and the TIM. The heatsink can vary in material composition, size, and geometry depending on the requirements of the design, such as power density, available air flow, and cost. The thermal resistance of a heatsink is highly variable with volume, material, air flow, baseplate thickness, and fin geometry. Key considerations in fin geometry include height, width, spacing, angle, and method of attachment to the baseplate [3]–[9]. For the

extruded fin, linear fins and baseplate are fabricated as one solid piece, which may limit the options for fin width and spacing [4]–[6].

The selection of heatsink geometry is partly determined by available air flow. Heatsink thermal resistance is inversely related to both surface area and air flow. Wider fin spacing may reduce overall surface area, but it may also improve ducted air flow in a forced-air configuration [5]. In a natural convection configuration without forced air from a fan, the thermal resistance is strongly related to the air flow generated by the thermal gradient across the length of each fin. The naturally produced air flow across the fins enables heat extraction through convection, which is also supplemented by radiation. For a natural convection thermal design, a vertically oriented pin-fin or cross-cut heatsink may provide the best thermal resistance for a given heatsink volume [8]. More advanced heatsink options include embedded heat pipes or liquid-cooled plates, which use internal fluid flow to improve the performance of the heatsink [7].

Between the baseplate of the heatsink and the mating surface of the heat source, a thermally conductive material is typically used to reduce the thermal resistance of the interface. This TIM may be a solid pad, a liquid gap filler, thermal grease, or a phase-change material. In many applications, the heatsink must be galvanically isolated from the converter, requiring a TIM with high electrical volume resistivity and dielectric breakdown voltage [10], [11]. A combination of solid and liquid TIMs may be used, for example, a pad with a thin layer of grease on each side, providing sufficient electrical isolation and mechanical support while also filling air gaps to reduce the thermal resistance.

Table 1 lists some examples of these TIMs. Most data sheets specify the material's thermal conductivity, which can be used to calculate the thermal resistance for a given volume of the material. The data sheet for a TIM pad may also specify the thermal resistance as a function of area and applied pressure. If specified, the thermal resistance also considers the thermal bond on both surfaces of the pad ("surface wetting"), as well as the impact of mechanical compression. Table 1 lists the specified thermal conductivity for each example, as well as the thermal resistance for a 10-mm² area and 1-mm thickness, uncompressed, and compressed with 50 psi. The electrical volume resistivity and dielectric breakdown voltage are also listed.

It is clear from this brief survey that a wide range of TIMs are available for different applications. Even within the options for silicone-based thermal pads, the resistance can vary by a factor of 50 depending on the properties of the material [12]–[15]. Other advanced solutions, such as phase-change materials, may be excellent options for certain applications, but some are not rated to withstand high voltage [16], [17].

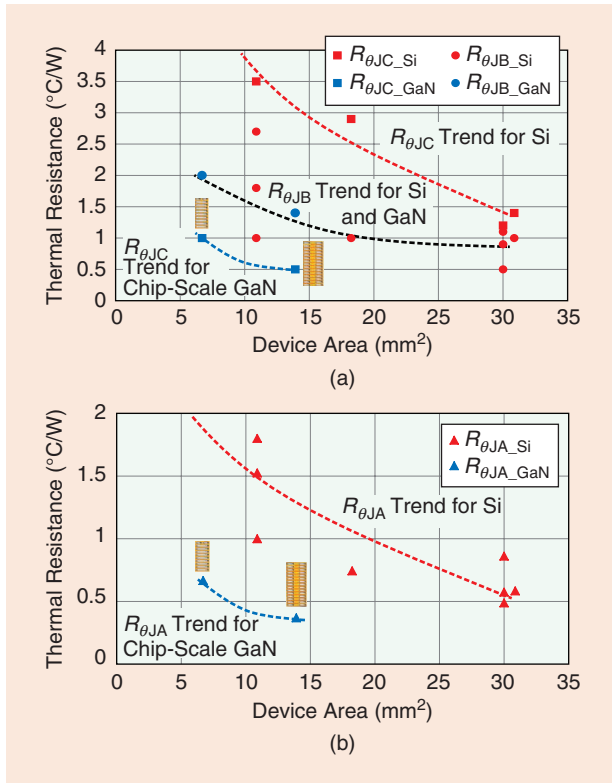


FIG 3 The thermal resistances for some typical LGA GaN transistors and packaged Si MOSFETs. (a) Specifications for junction-to-case (trend shown in black) and junction-to-board (trends shown in red and blue) and (b) minimum possible junction-to-ambient for combined case and board thermal paths for both GaN and MOSFETs [2]. (Source: EPC; used with permission.)

Table 1. Examples of commercially available TIMs.

Part Number	Form Factor	Thermal Conductivity	Thermal Resistance for 10-mm ² Area, 1-mm Thickness		Electrical Volume Resistivity	Dielectric Strength
			Uncompressed	50-psi Compression		
Bergquist Sil-pad 400 [12]	Silicone pad	0.9 W/m·K	Not specified	409 °C/W	10 ¹¹ Ω·m	20 kV/mm
Wakefield-Vette 120 [13]	Silicone grease	0.735 W/m·K	136 °C/W	N/A	5 × 10 ¹² Ω·m	9 kV/mm
Bergquist GF4000 [14]	Liquid gap filler	4 W/m·K	25 °C/W	N/A	10 ¹⁰ Ω·m	18 kV/mm
t-Global TG-X [15]	Ultrasoft silicone pad	12 W/m·K	19 °C/W	8 °C/W	10 ⁹ Ω·m	12 kV/mm
t-Global TG-S606P [16]	Silicone grease	8 W/m·K	12.5 °C/W	N/A	10 ¹² Ω·m	Not specified
Laird TPCM-580 [17]	Phase-change material	3.8 W/m·K	Not specified	0.8 °C/W	3 × 10 ¹⁰ Ω·m	Not specified

Heatsink Attachment for Bottom-Side Cooling

For traditional bottom-cooled [e.g., power quad flat no-lead (PQFN)] devices, the thermal resistance down through the PCB is the main heat flux path. This typically requires multiple thermal vias or a copper inlay, allowing improved heat flow through the PCB [18]. On the opposite side, a heatsink is often added to reduce PCB-to-ambient thermal resistance, as shown in Figure 4.

Bottom-side cooling presents several design challenges. First, the thermal vias or copper inlay may block the return path for the optimum vertical power loop layout [19]. Second, both thermal vias and copper inlays may require a costlier PCB. Third, the bottom side of the board is typically left unpopulated so that the heatsink can be mounted flush against the PCB. If components on the bottom side of the board interfere with heatsink attachment, the heatsink can be milled to accommodate them, or a thicker layer of TIM can be used. Either of these options will impact the cost and performance of the thermal design.

Heatsink Attachment for Multisided Cooling

For a chip-scale GaN transistor, heat can be extracted from the top, bottom, and even the sides of the device. Significant heat can still be extracted through the solder bars or solder bumps to the connected copper area and adjacent layers of the PCB. But in this scenario, the heatsink and TIM can be attached directly to the top side of the transistor case, as shown in Figure 5, providing a more direct path to remove the heat that avoids the added thermal resistance of PCB vias.

Typically, a top-mounted heatsink utilizes some form of spacer or shim between the heatsink and PCB, along with a soft TIM. These mechanical considerations are important to limit excessive force being transferred to the device, especially during the assembly process. The spacer determines the minimum distance between the

heatsink and the GaN devices. The spacer height should be kept to a minimum, but it must be tall enough to account for variation in the height and tilt of the transistors, spacers, and other nearby components, such as gate drivers and capacitors. Either a soft highly compressible thermal

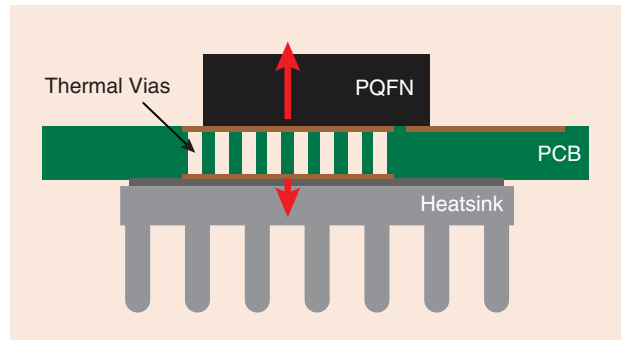


FIG 4 A cross-sectional diagram of a PQFN packaged device mounted on a PCB with a heatsink and thermal vias for bottom-side cooling. (Source: EPC; used with permission.)

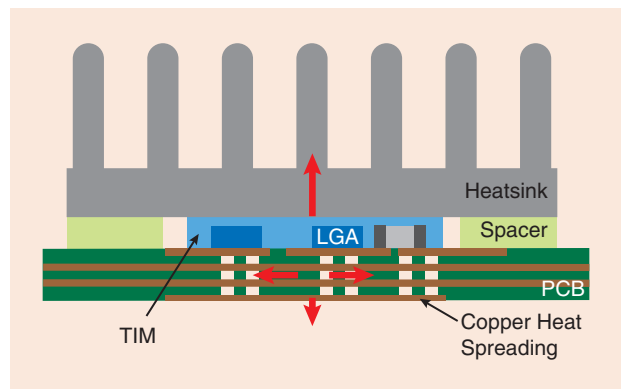


FIG 5 A cross-sectional diagram of a chip-scale LGA device mounted on a PCB with a heatsink attached for top-side or six-sided cooling. (Source: EPC; used with permission.)

pad or a liquid gap filler is recommended, as this limits mechanical stress and allows for a wider height tolerance between components.

Both top-side and bottom-side thermal paths should be optimized. First, the board-to-ambient thermal resistance can be reduced through the addition of thermal vias (similar to the bottom-cooled design) but with the thermal vias connecting to internal and external copper layers to spread the heat laterally. Second, the TIM resistance can be reduced in any of three ways:

- 1) by reducing the thickness of the device-to-heatsink interface
- 2) by selecting a highly thermally conductive TIM

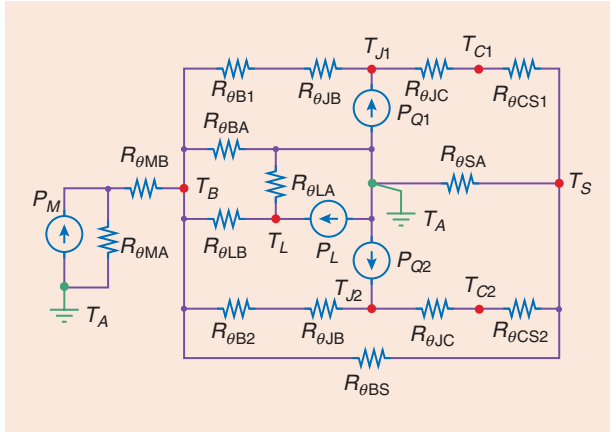


FIG 6 The thermal equivalent circuit for a half-bridge power stage utilizing top-side cooling, with power losses dissipated by two discrete GaN transistors, a filter inductor, and the surrounding system. (Source: EPC; used with permission.)

- 3) by placing the TIM on all sides of the devices and not just the top. This reduces the thermal resistance of $R_{\theta JC}$ and $R_{\theta CS}$, as the device perimeter side walls increase the overall device surface area by a significant margin. Liquid gap filler is a convenient option for achieving six-sided cooling in this manner.

Further improvements are possible by utilizing dual-sided heatsinking, forced-air cooling, liquid cooling, and through the use of exotic PCB materials, such as direct bonded copper [18], or insulated metal substrate [20].

Thermal Model of a Power Stage With Discrete GaN Transistors

The half-bridge circuit is a common building block in many power electronics circuits, and it will be used here as an example for thermally modeling a GaN-based power stage. A half-bridge circuit consists of two transistors, Q1 and Q2, as well as the supplementary components, for instance, gate drivers, regulators, and passive components [21]. The thermal behavior of a half-bridge power stage (e.g., a buck converter) using chip-scale GaN transistors can be modeled with the lumped-parameter equivalent circuit shown in Figure 6. This model includes parameters that represent most of the significant heat flux paths in the converter, as demonstrated in Figure 7. The three primary heat sources in this example system are power losses from Q1, Q2, and the filter inductor [21]. The lumped-parameter model neglects some of the more complex thermal interactions, such as treating the PCB temperature and heatsink temperature each as a single temperature node. In reality, a temperature gradient exists across each of these elements. Despite these simplifications, an equivalent circuit

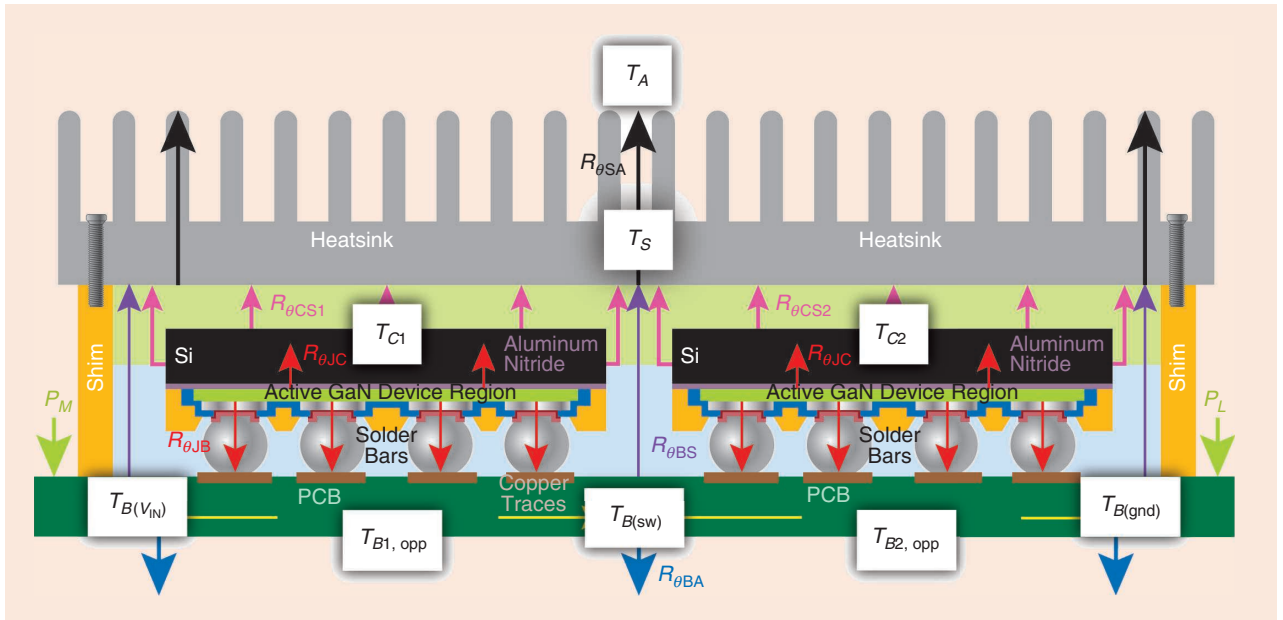


FIG 7 A cross-sectional diagram of a half-bridge circuit consisting of two chip-scale transistors and a heatsink, highlighting the physical heat flux paths from the two transistors corresponding to the equivalent circuit in Figure 6, as well as the heat contributions from the filter inductor and other system losses. (Source: EPC; used with permission.)

model can provide crucial insight into the thermal design of a system.

The temperature nodes in this model can be described as follows:

- T_{J1} and T_{J2} : The junction temperatures of Q1 and Q2, at the active GaN device region within each die
- T_{C1} and T_{C2} : The case temperatures of Q1 and Q2, at the top and sides of the substrate
- T_B : The board temperature, at a single location on the PCB copper close to both transistors; the exact location of this temperature depends partly on where it is measured, and this decision impacts the calculated thermal resistances
- T_S : The heatsink temperature, at a single location on the heatsink close to both transistors; similar to board temperature, this decision will impact the resulting thermal resistances in the model; most heatsinks specify the thermal resistance assuming a single heat source at the center of the baseplate, on the side opposite to the fins
- T_L : The filter inductor temperature
- T_A : The ambient temperature of the air surrounding the converter.

Likewise, the thermal resistances in this model represent the following physical heat paths (for a table of typical values, see [21]):

- $R_{\theta JC}$: Between the active GaN device region within each die to the exterior surface of the substrate
- $R_{\theta JB}$: Between the active GaN device region within each die to the PCB directly under the solder bumps/bars
- $R_{\theta B1}$ and $R_{\theta B2}$: Between the PCB directly under the solder bumps or solder bars of each transistor and the location on the PCB, designated T_B ; physically these resistances represent the heat spreading in the PCB near the solder joints where the thermal gradient is steep
- $R_{\theta CS1}$ and $R_{\theta CS2}$: Between the substrate of each transistor and the mating surface of the heatsink, through the TIM
- $R_{\theta BS}$: Between T_B and the mating surface of the heatsink, if any thermal path exists (e.g., if TIM is deposited across the entire PCB)
- $R_{\theta BA}$: Between T_B and ambient; this parameter models how well heat is spread throughout the PCB from solder

junctions, then transferred to ambient air through convection and radiation from the PCB surface area

- $R_{\theta SA}$: Between the heatsink and ambient; this parameter models how well the heatsink transfers heat to ambient air through convection and radiation
- $R_{\theta LB}$: Between the filter inductor and board, through the inductor's solder pads
- $R_{\theta LA}$: Between the filter inductor and ambient air, through the exposed surface of the inductor
- $R_{\theta MB}$ and $R_{\theta MA}$: These resistances account for thermal coupling through the PCB between the power stage and any other system losses.

Finally, the power losses are modeled as current sources, representing the following:

- P_{Q1} and P_{Q2} : Transistor power losses
- P_L : Filter inductor power losses
- P_M : If the power stage is integrated into a larger board design, this parameter accounts for any other sources of loss on the motherboard; in a discrete power stage, this could represent losses in the controller, gate driver, sensors, and regulators, although these contributions may be negligible in some designs.

Application Examples

A comprehensive methodology for mechanical attachment of a top-sided heatsink to a high-density buck converter was presented in [21]–[23], along with a detailed thermal characterization of the converter with and without the heatsink. Figure 8 demonstrates the procedure for assembling this thermal design example.

- 1) A thin nylon shim was cut to surround the power stage on three sides. The height of this shim was set at 1.02 mm, which is slightly taller than the transistors. Screw holes were drilled and tapped in the PCB to attach the shim and heatsink.
- 2) Liquid gap filler (GF4000) was deposited in the area enclosed by the shim.
- 3) An ultrasoft thermal pad (TG-X) was cut to the interior dimensions of the shim and attached to the bottom surface of the 10 × 15 × 15 mm cross-cut heatsink.
- 4) The heatsink was gently pressed down onto the nylon shim, positioned so that the heatsink mounted flush

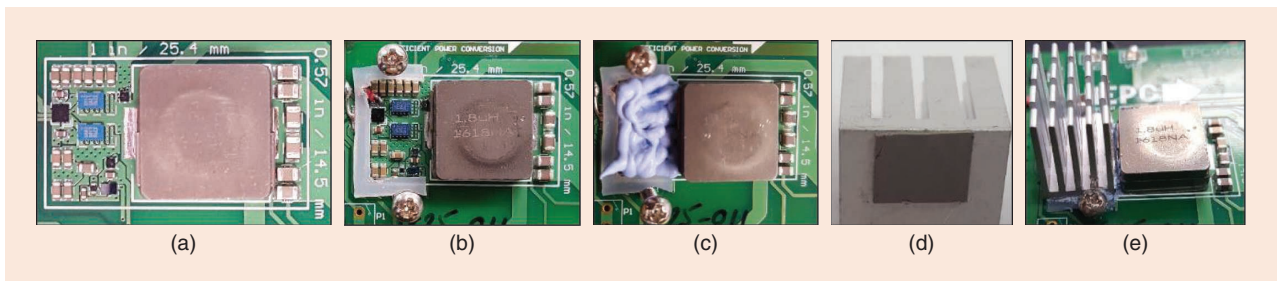


FIG 8 A heatsink attachment example for a high-density buck converter module using two EPC2045 GaN transistors, a nylon shim, TG-X thermal pad, GF4000 gap filler, 10 × 15 × 15-mm cross-cut heatsink, and screws for mechanical attachment. (a) PCB without heatsink, (b) nylon shim and attachment screws installed, (c) liquid gap filler deposited, (d) TIM pad attached to bottom of heatsink, and (e) final assembly [21], [22]. (Source: EPC; used with permission.)

against the shim, without any solid or liquid TIM between the heatsink and shim. The screws were tightened to attach the heatsink securely to the PCB, while the shim absorbed the mechanical stress of mounting. This step caused excess liquid gap filler to be extruded from beneath the heatsink, and this pressure forced the gap filler into air gaps across the uneven surface of the PCB.

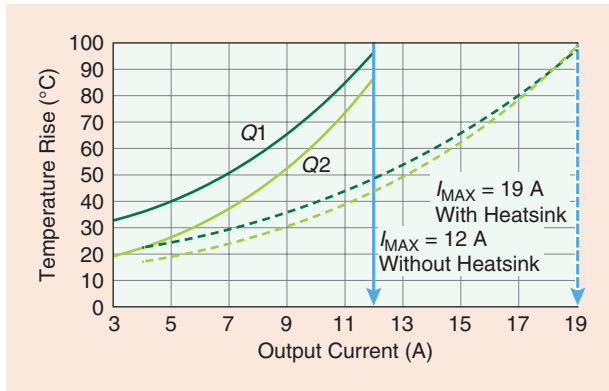


FIG 9 The transistor temperature rise for the example given in Figure 8, before and after heatsink attachment, with a cutoff of 100 °C for either transistor [21], [23]. (Source: EPC; used with permission.)

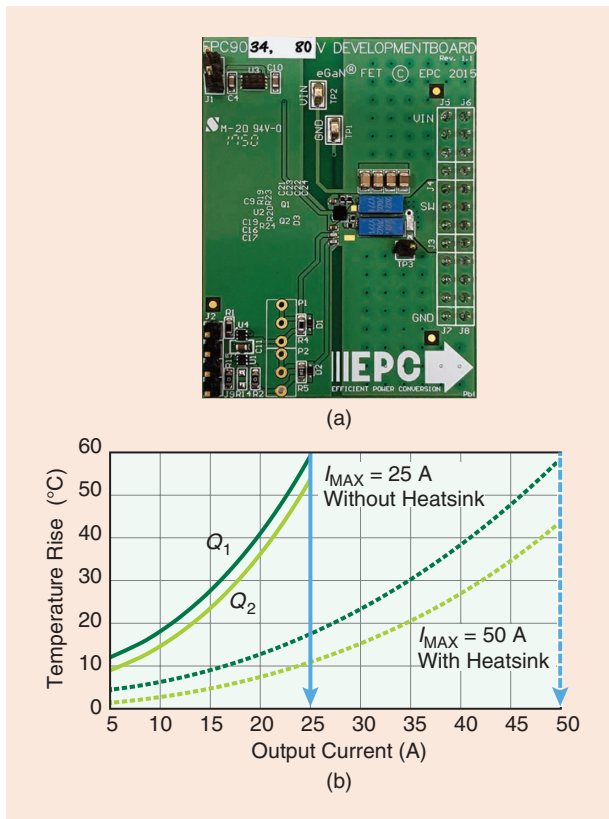


FIG 10 The heatsink attachment example using two EPC2206 GaN transistors, a nylon shim, TG-X thermal pad, and 45 × 45 × 25-mm cross-cut heatsink. (a) The PCB used for thermal evaluation and (b) transistor temperature rise before and after heatsink attachment, with a cutoff of 60 °C for either transistor [21]. (Source: EPC; used with permission.)

5) Excess gap filler was cleaned off the PCB. After several hours, the liquid gap filler cured to a solid state, and the assembly was complete.

Figure 9 shows the junction temperature rise of each EPC2045 GaN transistor in this example when operated as a 700-kHz, 48- to 12-V synchronous buck converter. The ambient temperature was maintained at 25 °C, with 800 linear feet per minute (LFM) forced air. Assuming a maximum allowable temperature rise of 100 °C above ambient for each transistor, the addition of the heatsink increased the converter's output current capability by 60%, from 12 to 19 A. Because of the close contact with the half-bridge circuit, the thermal impact of the Vishay IHLP series output filter inductor was also considered in this example [21].

As a second example, a 45 × 45 × 25-mm cross-cut heatsink was attached to a larger power stage using two EPC2206 GaN transistors and an off-board filter inductor. The heatsink was attached using only the TG-X thermal pad in this example, without liquid gap filler. Figure 10 shows this power stage and the junction temperature rise of each transistor, operated as a 125-kHz, 48- to 12-V synchronous buck converter. In this case, the maximum allowable temperature rise of each transistor was limited to only 60 °C above ambient, with 800 LFM forced air. The addition of the heatsink doubled the output current capability from 25 to 50 A [21].

Summary

This article reviewed the thermal management considerations for GaN transistors and GaN-based converter designs. The thermal characteristics typically included in device data sheets were explained, such as thermal resistances and transient thermal impedances. Thermal design schemes for bottom-side cooling and multisided cooling were discussed, including selection of the heatsink and TIM, as well as key design considerations for heatsink attachment.

Additionally, some thermal models for GaN transistor-based converters were defined, including an individual power stage with and without a heatsink. The typical options for temperature measurement were reviewed. Finally, this article gave two application examples for thermal design with chip-scale GaN transistors, from a single high-density buck converter to a much larger converter with higher voltage, more transistors, and a larger heatsink.

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